FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Advanced)-2019

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01– 06, 19 – 24, 37 - 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 - 46) *contains 12 questions. Based on this section contains* ***TWO (02)*** *paragraphs. Based on each paragraph, there are* ***TWO (02)*** *questions.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

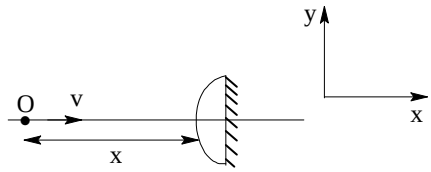
Physics

PART – I

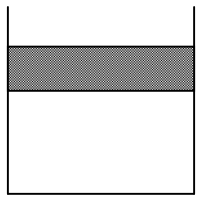
SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. When object 'O' moves towards a fixed lens mirror combination, select correct choice/choices:
(A) Image moves towards negative x-axis
(B) Speed of image and object may be same
(C) Image may move faster than object
(D) Image may come closer to arrangement



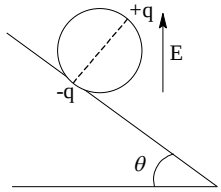
2. An ideal gas has molar heat capacity at constant pressure $C_p = 5R/2$. The gas is kept in a cylindrical vessel fitted with a piston which is free to move. Mass of the frictionless piston is 9 kg. Initial volume of the gas is $0.0027\ m^3$ and cross-section area of the piston is $0.09\ m^2$. The initial temperature of the gas is 300 K. Atmospheric pressure $P_0 = 1.05 \times 10^5 N/m^2$. An amount of $2.5 \times 10^4 J$ of heat energy is supplied to the gas such that piston move without acceleration, then:



(A) Initial pressure of the gas is 1.06×10^5 N/m^2
(B) Final temperature of the gas is 1000 K
(C) Final pressure of the gas is 1.06×10^5 N/m^2
(D) Work done by gas is 9.94×10^3 J

Space for Rough work

3. Two point charges $+q$ and $-q$ are fixed on diametrically opposite point on the uniform ring, such that $-q$ is at bottom and is in contact with a perfect insulator incline plane. Total mass is equal to m. It remains in equilibrium on the rough inclined plane in the presence of uniform vertical electric field $\vec{E}$. Then



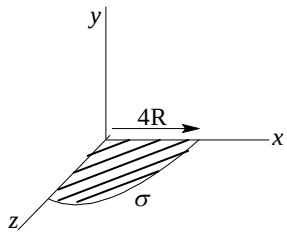
(A) The value of friction force is $mg\sin\theta$

(B) The value of friction force is less than $mg\sin\theta$

(C) The value of electric field is $\dfrac{mg}{2q}$

(D) The value of electric field is $\dfrac{mg\tan\theta}{2q}$

4. Quarter non–conducting disc of radius 4R having uniform surface charge density σ is placed in xz-plane then which of the following is the correct:



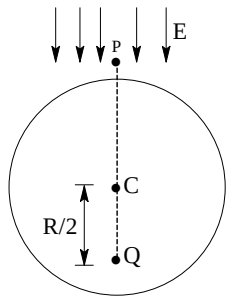
(A) Electric potential at (0, 3R, 0) is $\dfrac{\sigma R}{4\varepsilon_0}$

(B) Electric potential at (0,0,0) is $\dfrac{\sigma R}{2\varepsilon_0}$

(C) Electric field at (0, 3E, 0) is symmetric with x and z axis

(D) Electric field intensity at (-4R, 0, -4R) is equally inclined with x and z axis

5. A positive charge q is placed inside a neutral hollow conducting sphere of radius R, as shown in figure. Whole system is placed in uniform external vertical electric field pointing downward (line PCQ is also vertical) then select the correct statement(s) about electric field at point P. Point P is a point of the material inside the conductor.



(A) Electric field due to outer surface of sphere at point P is zero

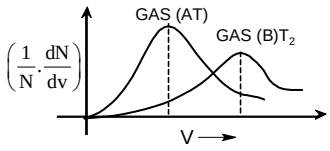
(B) Magnitude of electric field due to inner surface of sphere at point P is $q/9\pi\varepsilon_0 R^2$

(C) Magnitude of electric field at point P, due to hollow sphere is less than E

(D) Direction of resultant electric field at point P due to charge q and hollow sphere is vertically upward

Space for Rough work

6. According to Maxwell's distribution of molecular speeds, for the below graph drawn for two different samples of gases A and B at temperature T_1 and T_2 respectively, which of the following statements is/are INCORRECT :

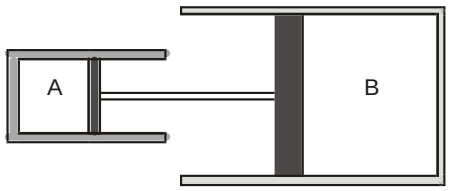


(A) If $T_1 = T_2$, then molecular mass of gas B (M_B) is greater than molecular mass of gas A (M_A).
(B) If molecular mass of gas A (M_A) is equal to molecular mass of gas B(M_B), then $T_1 > T_2$.
(C) If $T_1 < T_2$, then molecular mass of gas B(M_B) is necessarily less than molecular mass of gas A (M_A).
(D) If gas A is O_2 and gas B is N_2, then considering them to be ideal gases, T_1 is necessarily less than T_2.

(Paragraph Type)

This section contains **TWO** paragraphs. Based on each paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 7 and 8



Two fixed and horizontal cylinders A and B having pistons (both massless) of cross sectional area 100 cm^2 and 200 cm^2 respectively, are connected by massless rod. The piston can move freely without friction. The cylinder A contains 100 gms of an ideal gas (γ = 1.5) at pressure $10^5 N/m^2$ and temperature T_0. The cylinder B contains identical gas at same temperature T_0 but has different mass. The pistons are held at the state such that volume of gas in cylinder A and cylinder B are same and is equal to $10^{-2} m^3$. The walls and piston of cylinder A are thermally insulated where as gas in cylinder B is maintained at constant temperature T_0. The whole system is in vacuum. Now the pistons are slowly released and they move towards left and mechanical equilibrium is reached at the state when the volume of gas in cylinder A becomes $25 \times 10^{-4}\ m^3$.

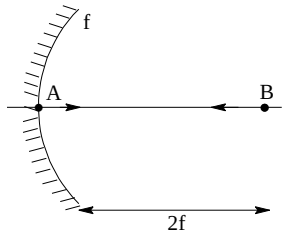
7. The mass of gas in cylinder B is
(A) 200 gms (B) 600 gms
(C) 500 gms (D) 1 kg

8. The change in internal energy of gas in cylinder A is :
(A) 2000 J (B) 1000 J
(C) 500 J (D) 3000 J

Space for Rough work

Paragraph for Question Nos. 9 and 10

At time t = 0, two point objects A and B respectively are at pole and centre of curvature of a fixed concave mirror of focal length f. The velocity vectors of A and B are $\vec{v}_A = u\hat{i}$ and $\vec{v}_B = -u\hat{i}$ respectively, where $\hat{i}$ is unit vector along principal axis directed from pole towards focus and u is a positive constant.



9. The distance between images of A and B will be 4f at time t =

(A) $\frac{f}{\sqrt{2}u}$ (B) $\frac{\sqrt{2}f}{u}$

(C) $\frac{f}{2u}$ (D) $\sqrt{2}u$

10. The magnitude of relative velocity of image of A and image of B at t = 0 is:

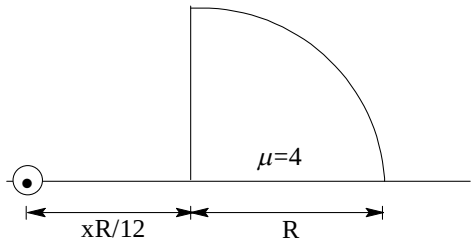
(A) u (B) 2u

(C) $\frac{u}{2}$ (D) $\sqrt{2}u$

SECTION – D
(Numerical Answer Type)

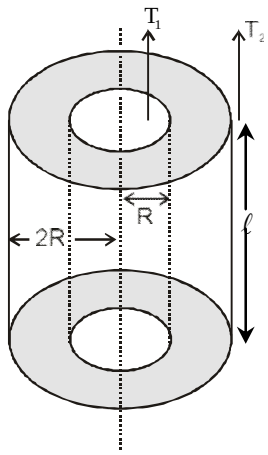
This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. Find the value of x so that image of object formed at infinite from a quarter cylinder of refractive index $\mu = 4$.



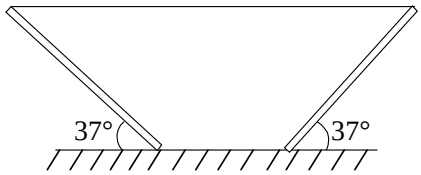
Space for Rough work

12. Inner surface of a cylindrical shell of length ℓ and of material of thermal conductivity k is kept at constant temperature T_1 and outer surface of the cylinder is kept at constant temperature T_2 such that $(T_1 > T_2)$ as shown in figure. Heat flows from inner surface to outer surface radially outward. Inner and outer radii of the shell are R and 2R respectively. Due to lack of space this cylinder has to be replaced by a smaller cylinder of length $\frac{\ell}{2}$ inner and outer radii $\frac{R}{4}$ and R respectively and thermal conductivity of material nk. If rate of radially outward heat flow remains same for same temperatures of inner and outer surface i.e. T_1 and T_2, then find the value of n.

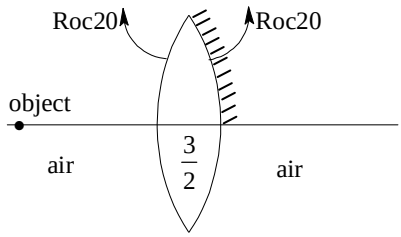


13. In a certain hypothetical radioactive decay process, species A decays into species B and species B decays
into species C according to the given reactions.
A→2B + particles + energy
B→3C + particles + energy
The decay constant for species A is $\lambda_1 = 1\ \text{sec}^{-1}$ and that for species B is $\lambda_2 = 100\ \text{sec}^{-1}$. Initially 10^4 moles of species of A were present, while there were none of B and C. It was found that species B reaches its maximum number at a time t_0 = 2 ln (10) sec. Calculate the value of maximum number of moles of B.

14. Two uniform identical rods of same mass are tied together with the help of a string and balanced as shown in the figure. The minimum coefficient of friction for which the system will remain in equilibrium in the position is $\frac{2}{x}$. The value of 'x' is

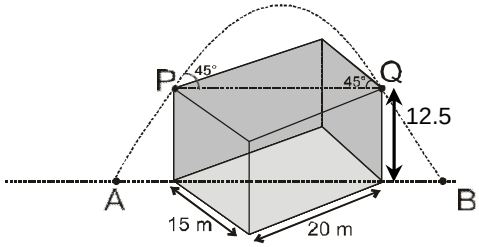


15. Distance between two images formed by upper and lower part of the point object placed at 30 cm from given lens is (60 + x) cm, then the value of x is:

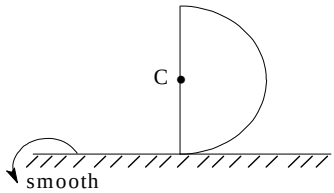


Space for Rough work

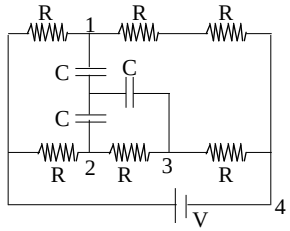
16. A commander fires a shell at certain angle of projection from 'A' which clears the building (cuboid) of dimensions 20 × 15 × 12.5 m^3 in its diagonal plane. Shell just clears the roof diagonally at points P and Q and falls on the other side of the building at B, and makes 45° angle with horizontal at P and Q as shown in the figure. If the range AB of the shell is $N\times\sqrt{3}$ metre then find N. (g = 10 m/s^2)



17. If angular acceleration of the uniform disc just after release from rest position shown in figure is $\frac{8\pi g}{NR}$, then calculate N. (C is centre of semi-circular disc.) ($\pi^2 = 10$)



18. In the show circuit, all three capacitor are identical and have capacitance $C\mu F$ each. Each resistor has resistance of $R\Omega$. An ideal cell of emf V volts is connected as shown. If the magnitude of potential difference across capacitor C_3 in steady state is $\frac{a}{b}V$ then value of $a\times b$ is:



Space for Rough work

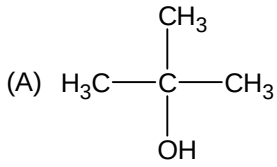
Chemistry

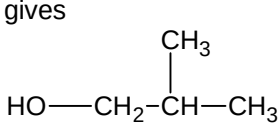
PART – II

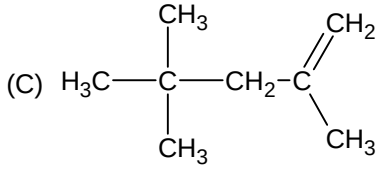
SECTION – A
(One or More than one correct type)

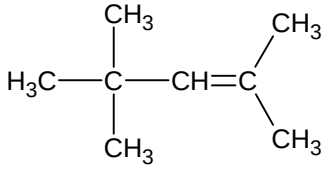
This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. The reaction of isobutene with conc.H_2SO_4 at 70^oC gives

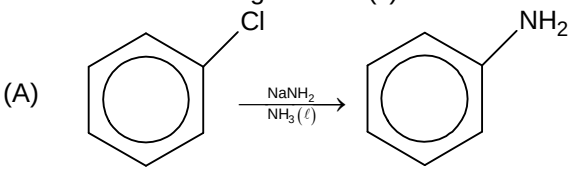
(A) 

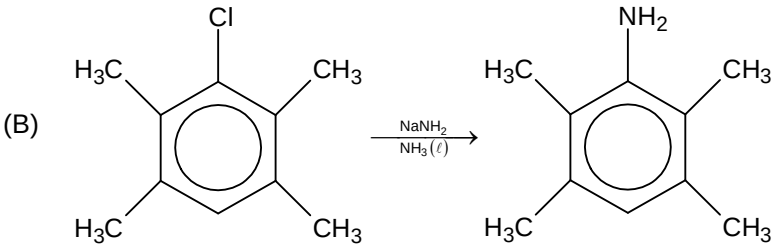
(B) 

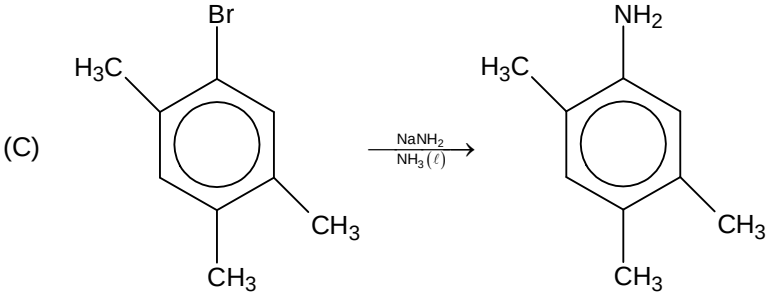
(C) 

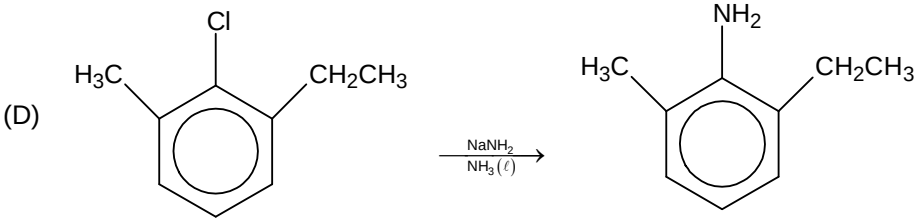
(D) 

20. Which of the following reaction(s) is/are not feasible?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

21. Example of step growth polymer is/are
(A) nylon-6 (B) nylon-6, 6
(C) polybutadiene (D) none of these

22. Which of the folloiwng alloys contain (s) Cu and Zn?
(A) Bronze (B) Brass
(C) Gun metal (D) Type metal

23. Which of the following statement is correct about the compound of Xe?
(A) $XePtF_6$ was first real compound of any of the noble gases
(B) with SbF_5, XeF_2 act as an Lewis acid but XeF_4 acts as a Lewis base
(C) with SbF_5 both XeF_2 and XeF_4 act as Lewis acid
(D) with SbF_5 both XeF_2 and XeF_4 act as Lewis base

24. Which of the following statements is false for $t_{1/2}$ (half life)?
(A) $t_{1/2}$ increases with increase in temperature
(B) $t_{1/2}$ decreases with increase in temperature
(C) $t_{1/2}$ is independent of temperature
(D) $t_{1/2}$ is the time taken when half the actual time is taken for the chemical reaction to complete

(Paragraph Type)

This section contains **TWO** paragraphs. Based on each paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 25 and 26

In a fuel cell, chemical enegy of fuel is used for producing electricity. H_2 is used as a fuel and oxygen is used as oxidiser. The standard enthalpy of combustion of hydorgen is -241 kcalmol^{-1}. The $E^{\circ}_{cell} = 1.23$ V

25. The equilibrium constant of the reaction is
(A) 4.7×10^{41}
(B) 3.7×10^{41}
(C) 4.210×10^{41}
(D) 4.169×10^{40}

Space for Rough work

26. The thermodynamic efficiency $\frac{\Delta G^o}{\Delta H^o} * 100$ of above cell is

(A) 95% (B) 70%
(C) 98.4% (D) 100%

Paragraph for Question Nos. 27 and 28

$t_{1/2}$ for n^{th} order reaction. If a reaction $nA \rightarrow$ Products

then $\text{rate} = \frac{-d[A]}{dt} = k_n[A]^n \Rightarrow \frac{-d[A]}{[A]^n} = k_n dt$

Integrating both sides and doing further reaction we get $t_{1/2} = \frac{2^{n-1}-1}{k_n(n-1)a_0^{n-1}}$

Where a_0 = initial conc. of A. Hence $t_{1/2} \propto \frac{1}{a_0^{n-1}}$.

27. For a reaction of nth order the ratio of $\frac{t_{1/2}}{t_{3/4}}$ is a function of

(A) a_0 (B) k_n
(C) n (D) t

28. The following data was obtained for a chemical reaction

a_0 (mol/dm^3)	0.5	1.1	2.48
$t_{1/2}$(s)	4280	885	174

The order of the reaction is

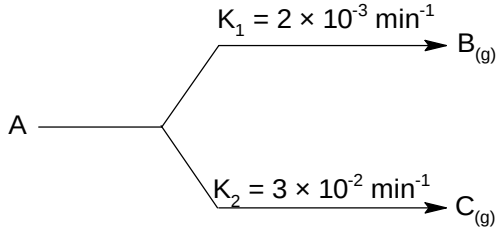
(A) 1 (B) 2
(C) 3 (D) 4

Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. A gaseous reactant A decomposes to produce gaseous products B and C in parallel reaction both by first order, as follows



If decomposition is carried out in a sealed flask, partial pressure of $B_{(g)}$ at the end of reaction was found to be 40 mm of Hg. Determine partial pressure of C after 1.0 hour.
(Given that antilog 0.8336 = 6.818)

30. The hydrogen atom in the ground state is excited by mass of monochromatic radiations of wavelength λ Å. The resulting spectrum consists of maximum 15 different lines. What is the value of λ in Å? $R_H = 109737$ cm^{-1}

31. A metal is known to form fluoride MF_2. When 10 ampere electricity is passed through a molten salt for 330 sec, 1.95 g metal is deposited. Find out the atomic weight of metal. What will be the quantity of charge required to deposit the same mass of Cu from $CuSO_4$(aq) in kilo Coulomb. (At. wt. of Cu = 63.6)

32. If boiling point of an aqueous solution is 100.1°C. What is its freezing point? Given latent heat of fusion and vaporization of water are 80 cal g^{-1} and 540 cal g^{-1} respectively. [Write the answer in the modulus form in °C]

33. What is resistance (in a cell with constant 0.206 cm^{-1}) of an 0.04 M solution of acetic acid at 25°C? Given, $K_a = 1.8 \times 10^{-5}$, $\wedge_m^o(CH_3COOH) = 390.5$ ohm^{-1} cm^2 mol^{-1}
[Write the answer in ohm]

Space for Rough work

34. Decomposition of phosgene gas

$COCl_2 \rightleftharpoons CO + Cl_2$ follows first order kinetics with and long

$$\log K_f = 13.48 - \frac{52400}{4.5} \times \frac{1}{T} \ldots \text{(i)}$$

$$\log K_b = 8.023 - \frac{26300}{4.5} \times \frac{1}{T} \ldots \text{(ii)}$$

Determine ΔS^0

[Write the answer in $J\text{-}K^{-1}$]

35. An ideal gas has a specific heat at constant pressure $C_p = \frac{5}{2}R$. The gas is kept in a closed vessel of volume 0.0083 m^3, at a temperature of 300 K and pressure 1.6×10^6 N/m^2. An amount of 2.49×10^4 J of energy is supplied to the gas. Calculate the final temperature of the gas in Kelvin.

36. In following organic reaction

$$\underset{(C_8H_{10})}{A} \xrightarrow{NBS} \underset{(C_8H_9Br)}{B} \xrightarrow{(CH_3)_3CO^-Na^+} \underset{(C_8H_8)}{C} \xrightarrow[(\text{reductive})]{\text{ozonolysis}} \underset{(C_7H_8O)}{D} + HCHO \xrightarrow[\substack{(iii)\,PCC \\ (iv)\,NH_2NH_2/OH^-}]{\substack{(i)\,CH_3MgBr \\ (ii)\,H_3O^+}} E$$

If the number of geometrical isomers of compound C is x and molecular weight of compound E is y then find the value of (x + y).

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Let $g(x) = f(\tan x) + f(\cot x)$ $\forall\ x \in \left(\frac{\pi}{2},\ \pi\right)$. If $f''(x) < 0$ $\forall\ x \in \left(\frac{\pi}{2},\ \pi\right)$ then

(A) g(x) is increasing in $\left(\frac{\pi}{2},\ \frac{3\pi}{4}\right)$

(B) g(x) is increasing in $\left(\frac{3\pi}{4},\ \pi\right)$

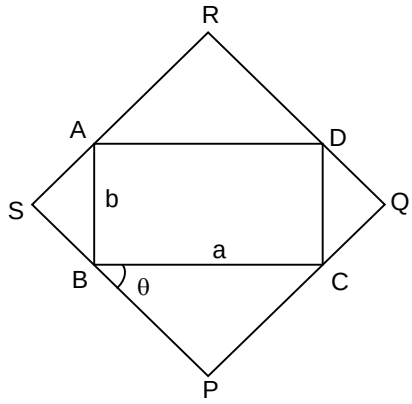
(C) g(x) is decreasing in $\left(\frac{3\pi}{4},\ \pi\right)$

(D) g(x) has local maximum at $x = \frac{3\pi}{4}$

38. The value of $\lim\limits_{x\to -4} \frac{\tan \pi x}{x+4} + \lim\limits_{x\to\infty}\left(1+\frac{1}{x^2}\right)^x$ is

(A) greater than 3
(B) greater than 4
(C) greater than 5
(D) less than 6

39. Let ABCD is a rectangle whose sides given are a and b. A rectangle PQRS whose area is A is shown in figure, then



(A) area A is maximum when $\theta = \frac{\pi}{3}$

(B) area A is maximum when $\theta = \frac{\pi}{4}$

(C) maximum value of area $A = \frac{1}{4}(a+b)^2$

(D) maximum value of area $A = \frac{1}{2}(a+b)^2$

Space for Rough work

40. The set S of all real x for which $(x^2 - x + 1)^{x-1} < 1$ contains
(A) (– 5, –1) (B) (– 1, 1)
(C) (– 1, 0) (D) (– 3, 1)

41. Let z be a complex number satisfying equation $z^n = \bar{z}^m$, where n, m $\in$ N, then
(A) if n = m, then number of solution of equation will be finite
(B) if n = m, then number of solution of equation will be infinite
(C) if n $\neq$ m, then number of solution of equation will be n + m
(D) if n $\neq$ m, then number of solution of equation will be n + m + 1

42. If u, v, w are real numbers such that $|ux + vy + wz| + |vx + wy + uz| + |wx + uy + vz| = |x + y + z| \ \forall$ x, y, z $\in$ R then
(A) u, v, w are equal (B) u, v, w are all of same sign
(C) u + v + w can take two values (D) |u| + |v| + |w| = 1

(Paragraph Type)

This section contains **TWO** paragraphs. Based on each paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 43 and 44

There is a point inside an equilateral triangle ABC of side d whose distance from the vertices is 3, 4, 5. Rotate the triangle and P through 60° about C. Let A go to A' and P to P'.

43. The area of triangle PA P' is
(A) 8 (B) 12
(C) 6 (D) none of these

44. The value of d^2 is
(A) $25 + 12\sqrt{3}$ (B) $25 - 12\sqrt{3}$
(C) $5 + 12\sqrt{3}$ (D) $25 + 24\sqrt{3}$

Space for Rough work

Paragraph for Question Nos. 45 and 46

Let P, Q are two points on the curve y = $\log_{1/2}(x-0.5)+\log_2\sqrt{4x^2-4x+1}$ and P is also on the circle $x^2 + y^2 = 10$, Q lies inside the given circle such that its abscissa is an integer.

45. The co–ordinates of P are given by
(A) (1, 2) (B) (2, 4)
(C) (3, 1) (D) (3, 5)

46. $\overrightarrow{OP}\cdot\overrightarrow{OQ}$, 'O' being the origin is
(A) 4 or 7 (B) 4 or 2
(C) 2 or 3 (D) 7 or 8

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. If the normal to the curve x = t – 1, $y = 3t^2 - 6$ at the point (1, 6) make intercepts a and b on x and y-axes respectively, then the value of $\frac{a+12b}{10}$ is ____

48. In a triangle ABC, side AB = 20, AC = 11, BC = 13, then the radius of the semicircle inscribed in triangle ABC, whose diameter lies on AB and is touching AC and BC is ___

49. g: R $\to$ R, if $g(x) = x^2 + x^2\int_{-1}^{1} tg(t)dt + x^3\int_{-1}^{1} g(t)dt$, then $11\int_{-1}^{1}(g(x)-g(-x))dx$ is _____

50. The figures 4, 5, 6, 7, 8 are written in every possible order. The number of numbers greater than 56000 is __

Space for Rough work

51. In the expansion of $(a^{1/3} + b^{1/9})^{6561}$, where a, b are distinct prime numbers, If the number of irrational terms are N then the value of $\frac{N}{100}$ is.

52. Let N = aaaaaa be a 6 digit number (all digit repeated) and N is divisible by 924 and let a, β be the roots of the equation $x^2 - 11x + \lambda = 0$, then product of all possible values of λ is ____

53. Let $2(1 + x^3)^{100} = \sum_{i=0}^{100}\left\{a_i x^i - \cos\frac{\pi}{2}(x+i)\right\}$, if $\sum_{i=0}^{50} a_{2i} = 2^k$, then the value of k is _____

54. Let f: [k, k + 1,…, 2007] $\to$ [1, 2, 3, …, n] be defined by $f(x) = \left[\frac{2007}{x}\right]$ (where [.] denotes the greatest integer function). The maximum value of k such that it is not possible to make f an onto function for any vale of n is.

Space for Rough work